Scale preservation is not enough: a profile-level independence countermodel for the Corollary 3.12 assembly inputs

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Abstract

We exhibit a concrete profile-valued countermodel showing that the remaining value-group profile switch in a conditional assembly from Theorem 3.11 to Corollary 3.12 in inter-universal Teichmüller theory is logically independent of the other assembly inputs. At $\ell = 7$, with labels $j \in \{1, 2, 3\}$ at one bad place and normalized unit $u = 1$, the theta-side readouts are $-j^2$ and the q-side readouts are -1, with equal procession weights $1/3$. The normalized generator and the unit-side readouts agree; membership, monotonicity, the hull domain and both volume decompositions hold. Nevertheless the aggregates are $-14/3$ and -1, the aggregate scale is forced to $3/14$, and the Corollary-3.12-shaped conclusion fails. For general odd primes $\ell \geq 5$ the aggregate scale is $12/(\ell(\ell+1))$. The countermodel and the resulting independence corollary are formalized in Lean 4; in the recorded selected build they depend only on the standard three axioms.

The profile and normalizations are not arbitrary choices. They are the source readouts: theta values $\underline{q}^{j^2}$, plain procession averaging, and the coefficient comparison in the proof of [IUTchIV, Theorem 1.10]. This note does not claim that Mochizuki's actual construction is this model, nor that Theorem 3.11, Corollary 3.12, the ABC conjecture or the switch H*val is false. It proves only that no derivation exists from the listed inputs once labelwise profile equality is removed. Further progress therefore requires a genuinely profile-level input, or a richer formalization of the actual construction, rather than more lemmas inside the same abstraction.

Keywords: inter-universal Teichmüller theory, Corollary 3.12, value-group profile, scale preservation, independence countermodel, Lean.

Note added in version 1.0.1. This version adds a remark in §2 relating the removed hypothesis (Hloc) to the Lean counterpart of H*val in [U26c], rewords the description of the Lean build, and rewrites the comments of the supplementary Lean files in English with the root namespace of the excerpts renamed. No theorem or conclusion is changed.

What is and is not claimed

Claimed.

- The hypotheses of the assembly capstone, with the single labelwise equality hypothesis removed, do not imply the Corollary-3.12-shaped volume conclusion. A concrete datum satisfies all remaining hypotheses and falsifies the conclusion.
- In this countermodel the chart/generator scale and the unit side are preserved; only the labelwise profile differs. The aggregate scale is nevertheless forced away from 1.
- The numerical profile, equal weights and q-side normalization used below are read off from the definitions and computations in IUT III–IV, as documented in Section 3.

Not claimed.

- That Mochizuki's actual Frobenioid construction reduces to the simple model in Section 4.
- That Theorem 3.11, Corollary 3.12, the ABC conjecture or H*val is false.
- That the whole assembly paper, the IUT black boxes, or frame fidelity have been formalized in Lean.

1 Scale versus profile

1.1 Two meanings of scale preservation

The phrase "the scale is preserved" has been used at two different layers in the assembly from Theorem 3.11 to Corollary 3.12. The first layer concerns calibration of the measuring device itself: a chart identifying two presentations has distortion $s_{\mathrm{chart}} = 1$, or a normalized generator has the same readout on both sides. The second layer concerns the aggregate obtained after summing labelwise components of the measured object with procession weights.

Fix one bad place and a positive normalized unit u. Write θ_j and q_j for the label-j theta-side and q-side readouts, and w_j for the weights. The aggregate ratio is

$$s_{\mathrm{agg}} := \frac{\sum_j w_j q_j}{\sum_j w_j \theta_j}.$$

Even when the chart and the weights are the same on both sides, s_{agg} need not equal 1 if the sequences $\{\theta_j\}_j$ and $\{q_j\}_j$ differ. We call such a sequence a *profile*. A profile is vector-valued information which becomes a scalar only after aggregation.

1.2 Relation to previous work

The note [U26] showed, by a countermodel, that Corollary 3.12 does not follow from the statements actually cited in its proof alone. The subsequent assembly note [U26c] used material outside that cited range to supply the scalar and unit layers at the statement level, while isolating the value-group profile switch H*val (there called R2V) as the central remaining input.

The present note connects these two results. The aggregate scale 3/14 in the countermodel of [U26] was described at the level of that cited theory as a scalar defect. At the level of concrete values it is forced by averaging the j^2 profile. Here the same phenomenon is placed inside the hypothesis shape of the assembly capstone of [U26c]. All supplied layers are satisfied, the profile alone differs, and the conclusion still fails. Thus the result is not a repetition of [U26]: it localizes the missing input inside the expanded assembly.

1.3 The question

We answer the following question in the negative: assuming a monotone volume, the hull domain condition, membership in the family of possible images, finite-sum decompositions on both sides, agreement at the normalized generator, and agreement of unit-side readouts, does the Corollary-3.12-shaped volume conclusion follow without labelwise profile equality?

2 The abstract assembly shape

We use the following abstraction of the capstone `cor312_of_assembly` in the Lean assembly development. Let Ω be a hull operator on a type α, let $\mathrm{Fam} \subseteq \mathcal{P}(\alpha)$ be a family of possible images, let $T \subseteq \alpha$, let $\mu : \mathcal{P}(\alpha) \to \mathbb{R}$ be a volume readout, and let $v_Q \in \mathbb{R}$. Let I be a finite index set with weights w_i and readouts θ_i, q_i. Consider the inputs

	input	in the model below
(A1)	μ is monotone	holds
(A2)	$\bigcup \mathrm{Fam}$ lies in the domain of Ω	holds
(A3)	$T \in \mathrm{Fam}$	holds
(A4)	$\mu(T) = \sum_{i \in I} w_i \theta_i$	both sides $-14/3$
(A5)	$v_Q = \sum_{i \in I} w_i q_i$	both sides -1
(G)	normalized generator readouts agree	both -1 at $j = 1$
(U)	unit-side readouts agree labelwise	both 0

The hypothesis deliberately removed is

$$\forall i \in I, \qquad q_i = \theta_i. \tag{Hloc}$$

The conclusion under test is the Corollary-3.12-shaped comparison

$$v_Q \leq \mu\Big(\Omega.\varphi\left(\bigcup \mathrm{Fam}\right)\Big). \tag{C}$$

In the usual assembly one obtains $\mu(T) \leq \mu(\text{hull})$ from membership and monotonicity (using extensivity of the hull operator), and $v_Q = \mu(T)$ from (Hloc), (A4) and (A5). The countermodel shows that (Hloc) is genuinely load-bearing in this inference.

Relation to the Lean counterpart of H*val. From version 1.0.2 of [U26c], the Lean counterpart of H*val is the pair of hypotheses `hmem`, which is (A3) here, and `hlocal`, which is (Hloc). The countermodel below keeps (A3) and removes only (Hloc). What it shows to be independent is therefore the profile part of H*val; the independence holds a fortiori for the pair.

The finite set I is arbitrary at this abstract level. Below we fix one bad place and take I to be its label set. Reading I instead as a set of places pre-aggregated over labels does not remove the mismatch: at the fixed bad place the two aggregates are already $-14/3$ and -1 (Section 7.2).

3 The source readout

An independence countermodel can be built from arbitrary sequences. What matters here is that the sequence, weights and normalization below are those read off from the source. This section lists only the inputs needed to fix the numbers; it is not a reproof of the source construction.

3.1 Labels and the theta pilot

For an odd prime ℓ, the label set is $\mathbb{F}_\ell^* = \mathbb{F}_\ell^\times/\{\pm 1\}$, with representatives $j \in \{1, \dots, \ell^*\}$, $\ell^* = (\ell-1)/2$. At a bad place the components of the theta pilot occur as the system of normalized theta values $\{\underline{\underline{q}}_{\underline{\underline{v}}}^{j^2}\}_j$ ([IUTchIII], Introduction, Definition 3.8(i), Remark 3.11.1(i)). Here $\underline{\underline{q}}$ is not the bare Tate parameter but the normalized generator of the next paragraph.

3.2 The q-side unit

[IUTchIV, Theorem 1.10] puts

$$\left|\log(\underline{\underline{q}})\right| = \frac{1}{2\ell}\log(\mathfrak{q}).$$

We write $u_{\underline{v}} := |\log(\underline{\underline{q}}_{\underline{\underline{v}}})| > 0$ and, with the sign convention for log-volumes, label readouts $-j^2 u_{\underline{v}}$ on the theta side and $-u_{\underline{v}}$ on the q side. In the main countermodel one bad place is fixed and $u_{\underline{v}}$ is normalized to 1.

3.3 Capsules and cancellation of the tensor factor

In the procession normalization in the proof of [IUTchIV, Theorem 1.10], one evaluates the components of the capsule $S^{\pm}_{j+1} = \{0, 1, \dots, j\}$ and then averages over $j \in \{1, \dots, \ell^*\}$. The distinguished factor is $i^\dagger = j$, and at a bad place the degree entering there is $\lambda = \mathrm{ord}(\underline{\underline{q}}_{\underline{\underline{v}}_j}^{j^2})$. A factor $j+1$ occurs in the per-factor quantities, but after the weighted average the capsule main term is $-\frac{j^2}{2\ell}\log(\mathfrak{q})$: the number of tensor factors cancels. The procession-averaged main term is $-\frac{\ell+1}{24}\log(\mathfrak{q})$. Comparing coefficients relative to $\log(\mathfrak{q})$ gives

$$\text{theta coefficient} = \frac{\ell+1}{24}, \qquad q \text{ coefficient} = \frac{1}{2\ell}, \qquad \frac{\text{theta}}{q} = \frac{\ell(\ell+1)}{12}.$$

The reciprocal aggregate scale is $12/(\ell(\ell+1))$; at $\ell = 7$ this is $3/14$, as in [U26].

3.4 Equal weights and the determinant

Procession normalization in [IUTchIII, Proposition 3.9(i)] is defined as the plain average over capsules. Remark 3.9.3 explicitly answers "no" to the question whether more general weighted averages may be used: compatibility with capsule inclusions forces the same weight for each capsule. Moreover $\det_{\otimes M}$ is normalized so as to reproduce the packet log-volume and may be omitted in volume-only arguments; the M-th tensor power comparison of (Ob4) is a common factor on both sides and does not change the profile ratio. The equal weights $1/3$ at $\ell = 7$ below are therefore the source definition, not a simplifying choice.

item	source location	value in the model
labels	$\ell^* = (\ell-1)/2$	$\ell = 7$, $\{1, 2, 3\}$
theta components	$\{\underline{\underline{q}}^{j^2}\}_j$	$-1, -4, -9$
q components	normalized triangle generator	$-1, -1, -1$
weights	Prop. 3.9(i), Rem. 3.9.3	$1/3, 1/3, 1/3$
aggregates	Thm. 1.10 main terms	$-14/3$ versus -1

4 The countermodel

We transcribe the definitions of the Lean file listed in Section 6.

Definition 4.1 (Profile datum at $\ell = 7$)**.** • Index set: `placesProf` $= \{1, 2, 3\}$.

- Weights: $w_j = 1/3$ (`wProf`).
- Theta readouts: $\theta_j = -j^2$ (`thetaVProf`).
- q readouts: $q_j = -1$ (`qVProf`).
- Unit readouts: 0 on both sides for every label (`unitThetaProf`, `unitQProf`).
- Carrier type $\alpha = \mathbb{R}$; hull operator Ω is the identity hull `trivialHull`: every subset is in the domain and is a hull, and $\Omega.\varphi(S) = S$.
- Theta region and family: $T = \{-14/3\}$ (`TProf`), Fam $= \{T\}$ (`FamProf`).
- Aggregate q value: $v_Q = -1$ (`vQProf`).
- Volume readout, for $S \subseteq \mathbb{R}$:

$$\mu(S) = \begin{cases} -\dfrac{14}{3}, & -\dfrac{14}{3} \in S, \\ -100, & -\dfrac{14}{3} \notin S \end{cases} \qquad (\texttt{muProf}).$$

The definitions of T and μ are intentionally minimal. They are not a reconstruction of the actual log-volume or holomorphic hull. Their purpose is to test what follows from the weak inputs listed in Section 2 without additional geometry of the actual hull. If the actual construction has stronger hull properties which rescue the conclusion, those properties are precisely an extra input beyond the list; the present independence result identifies that such an extra input is necessary.

5 Main theorems

Theorem 5.1 (Profile countermodel; Lean: `profile_countermodel`)**.** *The datum of Definition 4.1 satisfies all of the following.*

1. *μ is monotone, $\bigcup$Fam is in the domain of the identity hull, and $T \in$ Fam.*
2. *The decompositions*

$$\mu(T) = \sum_{j=1}^{3} \frac{1}{3}(-j^2) = -\frac{14}{3}, \qquad v_Q = \sum_{j=1}^{3} \frac{1}{3}(-1) = -1$$

   hold.

3. *At the normalized generator, $\theta_1 = q_1 = -1$, and the unit-side readouts agree at every label.*
4. *$v_Q = \frac{3}{14}\mu(T)$.*
5. *Nevertheless (Hloc) is false, and the conclusion $v_Q \le \mu(\Omega.\varphi(\bigcup \text{Fam}))$ is false.*

Proof. For monotonicity, let $A \subseteq B$. If $-14/3 \in A$ then $-14/3 \in B$ and both values are $-14/3$. If $-14/3 \notin A$, then $\mu(A) = -100$, while $\mu(B)$ is either $-14/3$ or -100; hence $\mu(A) \leq \mu(B)$. For the identity hull every set is in the domain and $\Omega.\varphi(S) = S$. Since $\mathrm{Fam} = \{T\}$, we have $T \in \mathrm{Fam}$ and $\bigcup \mathrm{Fam} = T$.

The finite sums are direct:

$$\sum_{j=1}^{3} \frac{1}{3}(-j^2) = -\frac{1+4+9}{3} = -\frac{14}{3}, \qquad \sum_{j=1}^{3} \frac{1}{3}(-1) = -1.$$

Also $-14/3 \in T$, so $\mu(T) = -14/3$. At the generator, $\theta_1 = -1^2 = -1 = q_1$, and both unit readouts are 0 by definition. The aggregate relation is $\frac{3}{14} \cdot (-14/3) = -1 = v_Q$.

On the other hand, at $j = 2$ we have $q_2 = -1$ and $\theta_2 = -4$, so (Hloc) fails. The conclusion would read $-1 = v_Q \leq \mu(\Omega.\varphi(T)) = \mu(T) = -14/3$, which is false. □

Corollary 5.2 (Aggregate scale; Lean: `aggregate_scale_eq`, `aggregate_scale_ne_one`)**.** *In the datum of Definition 4.1, any real number s satisfying $v_Q = s\,\mu(T)$ is equal to* $3/14$*. In particular $s \neq 1$.*

Proof. Solve $-1 = s(-14/3)$. □

Theorem 5.3 (Independence; Lean: `no_cor312_derivation_without_hlocal`)**.** *There is no universally quantified derivation of the conclusion (C) from inputs (A1)–(A5) alone. Precisely, it is false that for every hull operator Ω, family* Fam*, region T, readout μ, value v_Q, finite index set and weights and readouts, inputs (A1)–(A5) imply (C).*

Proof. Apply such a putative derivation to Definition 4.1. By Theorem 5.1 all hypotheses (A1)–(A5) hold while (C) is false, a contradiction. □

5.1 The computation for general ℓ

The formalized witness is at $\ell = 7$; one counterexample refutes the universal derivation. The aggregate computation is the same for every odd prime $\ell \geq 5$. Put $m = (\ell - 1)/2$, labels $j = 1, \ldots, m$, equal weights $1/m$, $\theta_j = -j^2 u$ and $q_j = -u$ with $u > 0$. Then

$$\frac{1}{m}\sum_{j=1}^{m} j^2 = \frac{(m+1)(2m+1)}{6} = \frac{\ell(\ell+1)}{12},$$

hence

$$s_{\mathrm{agg}} = \frac{-u}{-\frac{\ell(\ell+1)}{12}u} = \frac{12}{\ell(\ell+1)} \neq 1 \qquad (\ell \geq 5).$$

For example, at $\ell = 5, 7, 11, 13$ the mean profile coefficients are $5/2$, $14/3$, 11 and $91/6$. The general computation is the elementary finite-sum formula; the Lean theorem mechanizes the $\ell = 7$ witness.

5.2 Permutation-invariant readouts

Extracting only the label $j = 1$ would give the generator value $-u$. But a procession volume readout does not distinguish labels in this way. In general, a permutation-invariant linear readout must have equal weights: invariance under all transpositions forces all weights to coincide, and normalization on the constant profile $(1, \ldots, 1)$ forces each weight to be $1/m$. Such a readout is therefore the mean, which returns $\ell(\ell+1)/12 \neq 1$ on the theta side. The minimum readout does return 1, but it is not additive over tensor factors and hence is not a log-volume readout: for instance $\min(1,4) + \min(4,1) = 2$, whereas $\min((1,4) + (4,1)) = 5$.

6 Lean verification

The Lean module and namespace are

`M3.relative.ActualValuePartProfileIndependence`
`IUTFormal.M3.ActualValuePartProfileIndependence`

Its main declarations are `profile_countermodel`, `aggregate_scale_ne_one` and `no_cor312_derivation_without_hlocal`. The file and the module defining the hull operator were compiled with Lean 4.30.0 and Mathlib in the author's Lean development; the axioms reported for the main declarations were:

```
'...profile_countermodel' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'...aggregate_scale_ne_one' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'...no_cor312_derivation_without_hlocal' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

Thus there is no `sorryAx` and no additional axiom in the main theorems. The Lean file is recorded in commit `bc25c4c9` of the author's Lean development and is included in the supplementary package for this note.

Remark 6.1 (Scope of the build)**.** This selected build does not recompile the whole assembly development. The present file deliberately imports only the defining module of the `HullOperator` type also used by the assembly capstone, together with the needed Mathlib components. The hypothesis shape was matched by direct reading of `cor312_of_assembly` in `Cor312Assembly.lean`. What is machine-checked here is therefore the independence theorem of this note itself, not the assembly paper as a whole and not the actual IUT construction.

7 Objections and scope

7.1 The model is too degenerate

The model uses the identity hull and a singleton region. For an independence statement quantified over all data satisfying the listed properties, degenerate data are legitimate counterexamples. The actual holomorphic hull may of course have stronger closure, convexity or analytic properties which rescue the conclusion. In that case the rescuing property is exactly an additional input. The present result certifies that no derivation exists without such an addition.

7.2 The indices should be places, not labels

The finite sum in the capstone shape can be read with place indices or with combined (place, label) indices. We use the finer reading to expose the profile. Pre-aggregation at a fixed bad place does not help: the two place aggregates are already $-14u/3$ and $-u$. Across several bad places the same ratio $\ell(\ell+1)/12$ occurs at each place, and summing with coefficients $u_{\underline{v}} \geq 0$ produces no cancellation. Coarsening the indices or passing to a global sum therefore does not remove the mismatch.

7.3 The readout μ is not the actual log-volume

Correct, and stated as such above. Inputs (A1)–(A5) impose on μ only monotonicity and two decompositions. Additional laws satisfied by the actual log-volume — its full behaviour under tensor products, lower bounds coming from the spread of the hull, or detailed normalizations across places — are not in the input list. An argument using such laws may evade the countermodel, but then the additional law carrying the weight must be named. The point of this note is that one cannot say "the remaining lemmas will close the gap" without that naming.

7.4 Why not select the generator $j = 1$?

The two sides do agree at the normalized generator $j = 1$. That is chart-scale preservation. But the Corollary-3.12-shaped conclusion measures the hull of the whole family of possible images with procession-normalized volume, not a single generator. By Section 5.2, a permutation-invariant linear aggregation is necessarily the mean. An operation selecting one label asymmetrically is not part of the mono-analytic procession vocabulary; if additional structure justifies such a selection, it forms part of the profile switch and should be stated as such.

7.5 Why not assume aggregate scale $s = 1$ separately?

Assuming $s_{\mathrm{agg}} = 1$ removes the defect by fiat, but under the source readout the two aggregates have ratio $\ell(\ell+1)/12$, so $s_{\mathrm{agg}} = 1$ is incompatible with the readout itself. The mathematical question is how the theta-side quantity, before aggregation, is connected through the hull of possible images to the q-side quantity. Setting a numerical ratio to 1 afterwards does not prove that connection.

7.6 The actual construction may contain more

This is compatible with the present conclusion. For example, [EtTh] contains a local skeleton in which the difference of two sections is valued in the cyclotomic kernel and the theta class is pinned modulo units. The actual Frobenioid construction has far more structure than the abstraction of Section 2. What the countermodel shows is that at least one piece of that structure must carry weight at the profile level. Conversely, once a candidate additional mechanism is proposed, the countermodel indicates which listed hypothesis it must violate or strengthen: for instance the actual hull must spread strictly more than the identity hull, the actual family of possible images must be larger, or an object-level identification must change the labelwise readouts.

8 Conclusion

Preservation of chart calibration and of the normalized generator scale does not imply preservation after aggregation. When the measured profile changes from $\{j^2u\}_j$ to $\{u\}_j$, the same ruler and the same equal weights produce aggregates differing by the factor $\ell(\ell+1)/12$. The countermodel satisfies every other assembly input and still falsifies the conclusion. The missing input is therefore localized exactly at the profile switch, not distributed over unspecified "remaining lemmas".

From the point of view of formalization, this is a boundary result rather than a failure of effort. A method which stacks further lemmas inside the fixed abstraction cannot cross the boundary. Progress requires either formalizing the actual construction in richer definitions and exhibiting where the profile-level identification occurs, or explicitly introducing an object-level input equivalent to that identification. This note fixes that necessity in finite, machine-checkable form.

residual	content	relation to this note
R1	IUT black boxes	not verified here
R2V	value-group switch H*val	its profile part (Hloc) shown independent
R2	inhabitation / frame fidelity	no inhabitation claim is made
R3	independent re-verification	source audit and rereading remain

Disclosure

This work was carried out by the author as an independent researcher and is unrelated to any institutional affiliation. Generative AI was used under the author's direction in the audit, formalization and preparation of this draft. The author bears responsibility for its contents. This is an unrefereed preprint.

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